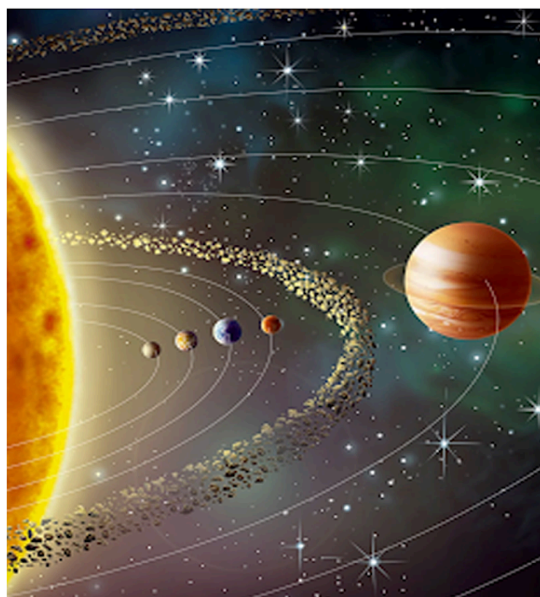


General Knowledge

World Geography

STUDY NOTES

Climatology: Atmospheric Circulation



Climatology: Atmospheric Circulation

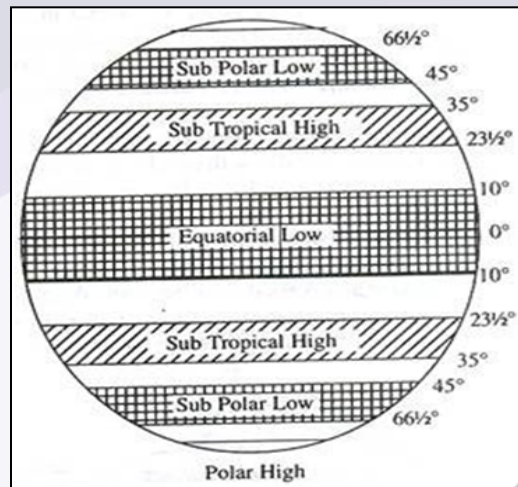
Atmospheric Pressure

- The weight of a column of air contained in a unit area from the mean sea level to the top of the atmosphere is called the atmospheric pressure.
- At sea level, the average atmospheric pressure is **1,013.2 millibar**.
- Due to gravity, the air at the surface is denser and hence has higher pressure.
- Air pressure is measured with the help of a mercury barometer or the aneroid barometer.
- **Pressure decreases with height.** At any elevation, it varies from place to place and its variation is the primary cause of air motion, i.e. wind which moves from high-pressure areas to low-pressure areas.

Vertical variation of pressure

- In the lower atmosphere, the **pressure decreases rapidly with height**.
- The decrease amounts to about 1 mb for each 10 m increase in elevation. However, it does not always decrease at the same rate.
- The **vertical pressure gradient force is much larger than the horizontal pressure gradient**.
- However, it is generally balanced by a nearly equal but opposite gravitational force. Hence, we do not experience strong upward winds.
- Small differences in pressure are highly significant in terms of the wind direction and velocity.
- **Isobars** are lines connecting **places having equal pressure**.

World Distribution of Sea Level Pressure



Equatorial low-pressure belt:

- Near the equator the sea level pressure is low and the area is known as **equatorial low**.
- This belt is thermally produced due to heating by the sun. Due to excessive heating horizontal movement of air is absent here and only vertical currents are experienced in this belt.
- Therefore, this belt is called the **doldrums (the zone of calm)**.
- This belt is also known as **Inter Tropical Convergence Zone (ITCZ)** because the trade winds flowing from the subtropical high-pressure belts converge here.

Subtropical high-pressure belt:

- Along 30° N and 30° S are found in the high-pressure areas known as the **subtropical highs**.
- The existence of these pressure belts is due to the fact that the rising air of the equatorial region is **deflected towards the poles due to the earth's rotation**.
- After becoming cold and heavy, it descends in these regions and results in high pressure.
- Calm conditions with feeble and variable winds are found here.
- In the southern hemisphere, this belt is broken by small low-pressure areas in summer over Australia and South Africa.
- In the northern hemisphere, the belt is more discontinuous by the presence of land masses, and high pressure occurs only over the ocean areas as discrete cells; these are termed the **Azores and Hawaiian cells** in the **Atlantic and Pacific areas** respectively.

- These belts are also called **Horse latitudes**. In older days, vessels with cargo of horses passing through these belts found difficulty in sailing under these calm conditions. **They used to throw the horses into the sea in order to make the vessels lighter.**
- In the upper atmosphere over this belt, the upper-level westerlies and anti-trade winds converge and set up **descending currents** in the atmosphere.

Sub-polar low pressure belt:

- Further polewards along 60°N and 60°S, the low-pressure belts are termed as the **subpolar lows**.
- These belts are **not thermally induced**; instead, the winds coming from the subtropics and the polar regions converge in this belt and rise upward.
- The great temperature contrast between the subtropical and the polar regions gives rise to **cyclonic storms** in this belt.
- In the southern hemisphere, this low-pressure belt is more pronounced due to the vast presence of the ocean and is referred to as the **sub-antarctic low**.
- But in the northern hemisphere, there are large land masses which are very cold. Therefore, the pressure over these landmasses is increased and the continuity of the belt is broken.

Polar high-pressure belt:

- Because of low temperatures, air compresses, and its density increases, hence, high pressure is found here throughout the year.
- This is more marked over the land area of the Antarctic continent than over the ocean of the North Pole.

These pressure belts are not permanent. They **oscillate with the apparent movement of the sun**. In the northern hemisphere in winter they move southwards and in the summer northwards. The most profound effect of shifting of belts is seen in the **temperate region**.

- Winds blowing from the Horse latitudes in the form of westerlies create unique climatic conditions in the temperate parts of the world, especially in the Mediterranean region.

Factors affecting the velocity and direction of the wind

- The air in motion is called wind. The wind blows from high pressure to low pressure.

- The **wind at the surface experiences friction**. In addition, the rotation of the earth also affects the wind movement.
- The horizontal winds near the earth's surface respond to the combined effect of **three forces** the pressure gradient force, the frictional force, and the Coriolis force. In addition, the gravitational force acts downward.

Pressure Gradient Force:

- The **difference in atmospheric pressure** produces a force.
- The rate of change of pressure with respect to distance is the pressure gradient.
- The pressure gradient is **strong where the isobars are close** to each other and weak where the isobars are apart.

Frictional Force:

- It affects the speed of the wind. It is **greatest at the surface** and its influence generally extends up to an elevation of 1 - 3 km.
- Over the sea surface, the friction is minimal.

Coriolis Force:

- The force exerted by the **rotation of the earth** is known as the Coriolis force.
- It deflects the wind to the **right direction in the northern hemisphere** and to the **left in the southern hemisphere**. The deflection is greater when the wind velocity is high.
- The Coriolis force is **directly proportional to the angle of latitude**. It is maximum at the poles and is absent at the equator.
- The Coriolis force acts **perpendicular to the pressure gradient force**.
- The pressure gradient force is perpendicular to an isobar. The higher the pressure gradient force, the higher the velocity of the wind and the larger is the deflection in the direction of the wind.
- As a result of these two forces operating perpendicular to each other, in the low-pressure areas, the wind blows around it.
- At the **equator, the Coriolis force is zero** and the wind blows perpendicular to the isobars. The low pressure gets filled instead of getting intensified. That is the reason why **tropical cyclones are not formed near the equator**.

